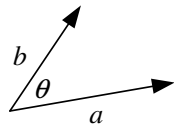


§12.3 The Dot Product

Homework : 11,17,19,23,25,27,31,37,39,51

Dot product (a scalar) Let $a = \langle a_1, a_2, a_3 \rangle$, $b = \langle b_1, b_2, b_3 \rangle$.

(1) 代數意義： $a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$

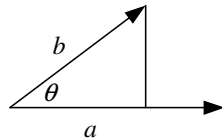


幾何意義： $a \cdot b = |a||b|\cos\theta$

(2) $a \perp b \Leftrightarrow a \cdot b = 0$.

(3) $\cos\theta = \frac{a \cdot b}{|a||b|}$.

(4) Scalar projection of b onto $a = |b|\cos\theta = \frac{a \cdot b}{|a|}$.



(5) Vector projection of b onto $a = \frac{a \cdot b}{|a|} \frac{a}{|a|} = \left(\frac{a \cdot b}{|a|^2} \right) a$.

(6) $a = \alpha b$ for some $\alpha \in \mathbb{R} \Leftrightarrow a \parallel b$

Example 1 : Find the scalar and vector projection of $b = \langle 1, 1, 2 \rangle$ onto $a = \langle -2, 3, -1 \rangle$.

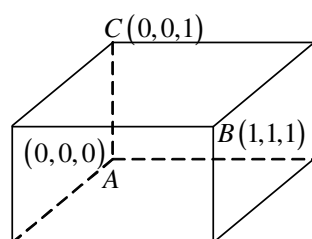
Solution :

$$\frac{a \cdot b}{|a|} = \frac{-2 + 3 - 2}{\sqrt{14}} = \frac{-1}{\sqrt{14}}$$

$$\frac{a \cdot b}{|a|} \frac{a}{|a|} = -\frac{1}{\sqrt{14}} \frac{\langle -2, 3, -1 \rangle}{\sqrt{14}} = \left\langle \frac{1}{7}, \frac{-3}{14}, \frac{1}{14} \right\rangle.$$

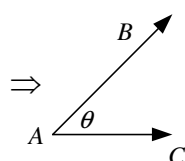
Example 2 : Find the angle between a diagonal of a cube and one of its edges.

Solution :



$$\overrightarrow{AC} = \langle 0, 0, 1 \rangle$$

$$\overrightarrow{AB} = \langle 1, 1, 1 \rangle$$



$$\Rightarrow \cos \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \cos^{-1} \frac{1}{\sqrt{3}} .$$